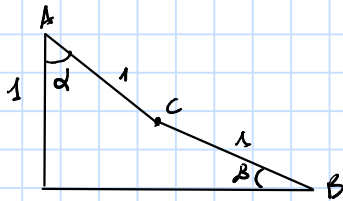


I° tratto $\rightarrow a_1 = g \cos \alpha$ e in moto uniformemente accelerato

II° tratto $\rightarrow a_2 = g \sin \beta$

$$a_2 = g - g \cos \alpha = g - a_1$$

cercare di esprimere il tempo totale $T = t_1 + t_2 = T(a_1)$ in funzione di a_1



$$T = t_1 + t_2$$

$$x(t_1) = \frac{1}{2} a_1 t_1^2 \rightarrow l = \frac{1}{2} a_1 t_1^2$$

$$x(t_2) = \frac{1}{2} a_2 t_2^2 = \frac{1}{2} (g - a_1) t_2^2 \rightarrow l = \frac{1}{2} (g - a_1) t_2^2$$

$$\begin{cases} t_1 = \sqrt{\frac{2l}{a_1}} \\ l = \frac{1}{2} (g - a_1) t_2^2 \end{cases}$$

$$\begin{cases} t_1 = \sqrt{\frac{2l}{a_1}} \\ \sqrt{\frac{2l}{g - a_1}} = t_2 \end{cases}$$

$$T = t_1 + t_2 = \sqrt{\frac{2l}{a_1}} + \sqrt{\frac{2l}{g - a_1}} = \sqrt{\frac{2l}{g \cos \alpha}} + \sqrt{\frac{2l}{g(1 - \cos \alpha)}}$$

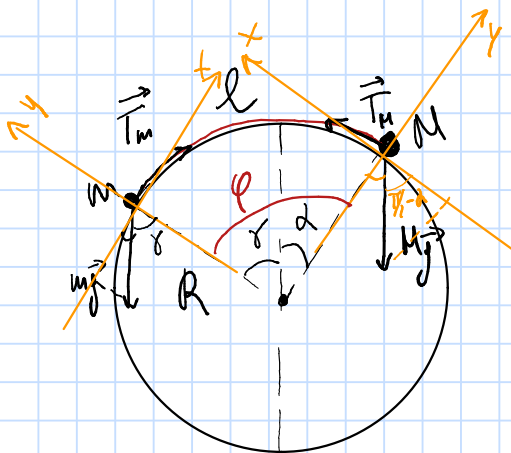
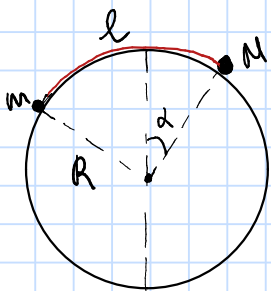
$$3.14 \text{ cm}$$

$$l = 3.14 \text{ mm}$$

$$R = 20 \text{ cm}$$

$$\alpha = 30^\circ \text{ eq. statico}$$

stimare il rapporto tra le masse



$$\begin{cases} M\vec{g} + \vec{T}_M = 0 \\ m\vec{g} + \vec{T}_m = 0 \end{cases}$$

$$\begin{aligned} \vec{T}_M &= -\vec{T}_m \\ T_M &= T_m \end{aligned}$$

$$Mg \sin \alpha + T_M = 0$$

$$-Mg \sin \alpha + T_m = 0$$

$$m g \sin \gamma + T_m = 0$$

$$m g \sin \left(\frac{\pi}{2} - \alpha \right) \rightarrow m g \cos(\alpha) + T_m = 0$$

$$l = R \cdot \varphi$$

$$3.14 \cdot 10^{-3} = 20 \cdot \varphi \rightarrow \varphi = \frac{\pi \cdot 20}{220}$$

$$\frac{\pi}{2} : \pi = x : 180 \rightarrow \frac{180 \cdot \frac{\pi}{2}}{\pi} = 90 \quad \gamma = 60^\circ$$

$$-Mg \sin \alpha + T_m = -m g \cos(\alpha) + T_m$$

$$Mg \sin \alpha = m g \cos \alpha$$

$$\frac{M}{m} = \frac{g \cos \alpha}{g \sin \alpha} \rightarrow \frac{M}{m} = \frac{1}{\tan \alpha}$$

$$Mg \sin \alpha = m g \sin \gamma$$

$$\frac{M}{m} = \frac{g \sin \gamma}{g \sin \alpha} \rightarrow \frac{M}{m} = \frac{\sin \left(\frac{\pi}{3} \right)}{\sin \left(\frac{\pi}{6} \right)} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$$